1 A student investigates stationary sound waves in cylindrical tubes. Fig. 1.1 shows a stationary wave pattern in a tube which is open at both ends.

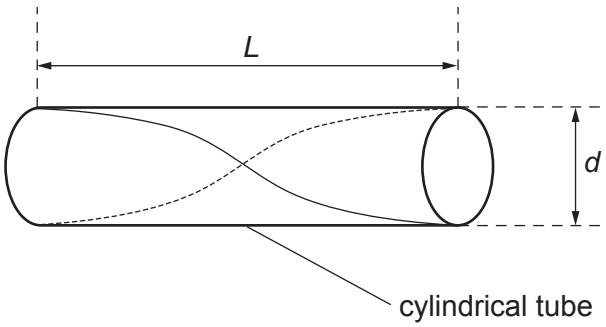

Fig. 1.1

The tube has length L and diameter d. The frequency of the sound for the stationary wave pattern shown is f.

There are a number of different tubes available.

It is suggested that the relationship between f and d is

$$\frac{v}{f} = 2L + kd$$

where v is the speed of sound in air and k is a constant.

Design a laboratory experiment to test the relationship between f and d.
Explain how your results could be used to determine values for k and v.

You should draw a diagram, on page 3, showing the arrangement of your equipment. In your account you should pay particular attention to:

- the procedure to be followed
- the measurements to be taken
- the control of variables
- the analysis of the data
- any safety precautions to be taken.

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